An Indoor Mixed-Reality Testbed for Multi-Drone Experiments with Real and Simulated Agents

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Abstract:
This paper presents a mixed-reality indoor testbed for multi-drone experiments with real and simulated agents. The proposed testbed extends our previous mixed-reality indoor arena, which supported only real or simulated drones at a time, by enabling their simultaneous coexistence within the same experimental environment. The setup integrates Crazyflie drones, an OptiTrack motion capture system for pose measurement of the real agents, an object-oriented MATLAB framework for dynamic simulation and implementation of guidance and control algorithms, a Python-based communication interface for command transmission to the real drones, and a 3D visualizer, developed in Unity 3D, for integrated visualization of all agents. In the proposed architecture, the poses of the real drones are acquired via the motion capture system, while the states of the simulated drones are computed via dynamic simulation based on the Crazyflie dynamic model. The testbed is demonstrated through three representative scenarios: a fully simulated case with three drones, a hybrid case with one real and two simulated drones, and a hybrid case with two real and one simulated drone. The results confirm the feasibility and flexibility of the proposed testbed for mixed-reality multi-agent experiments.

Keywords: autonomous drones; guidance and control; indoor flight experiments; mixed reality.

1. INTRODUCTION

Autonomous aerial systems have attracted significant interest, especially in cooperative and multi-agent applications. In this context, the development and validation of guidance and control algorithms require experimental frameworks that are both accessible and flexible (Okaya et al., 2023). Among the available platforms, the Crazyflie drone (Bitcraze, 2013) has become a suitable option for indoor research due to its compact size, open-source architecture, and suitability for laboratory experimentation.

Recent studies have employed the Crazyflie platform to evaluate guidance and control strategies under both numerical and experimental conditions. For example, Luis et al. (2020) have proposed a model predictive control (MPC) strategy for multi-robot trajectory tracking, validating the method in MATLAB and then implementing it experimentally in C++ using Crazyflie drones. Similarly, Soria et al. (2022) have employed an MPC strategy for aerial swarms, combining numerical evaluation in MATLAB with experimental implementation using Python. In another line of work, Giernacki et al. (2017) have proposed an integrated architecture for drone control development, combining a MATLAB/Simulink-based dynamic model with a three-dimensional visualization environment. More recently, Cruz et al. (2024) have introduced a mixed-reality indoor testbed that unifies numerical and experimental evaluation by combining real or simulated drones with virtual obstacles in a common environment. Although this testbed represents an important advance toward flexible mixed-reality experimentation, it is limited to the inclusion of virtual obstacles and does not support simultaneous coexistence of simulated and real drones in the same experimental enviroment.

To overcome this limitation, this paper extends the mixed-reality indoor testbed previously developed by Cruz et al. (2024) to support multi-drone experimental evaluation using both real and simulated aerial agents. The proposed testbed architecture integrates an OptiTrack motion capture system for pose feedback, a Python-based communication interface for the Crazyflie platform, an object-oriented MATLAB module for guidance, control, and simulation routines based on the Crazyflie dynamic model, and a photorealistic three-dimensional visualizer developed in Unity 3D. Unlike the previous version of the testbed (Cruz et al., 2024), which achieves mixed reality by integrating virtual obstacles with real drones, the proposed extension additionally enables the coexistence of simulated and real drones within the same environment. Consequently, the setup offers three modes of operation: a pure simulation mode, in which all entities are simulated; an experimental mode, which integrates real drones with virtual obstacles; and a mixed mode, which combines both real and simulated drones alongside virtual obstacles within the same environment. These operation modes provide evaluation flexibility, while the inclusion of Crazyflie dynamics in the simulation model enhances predictability

⋆ Project developed at the Air Robotics Laboratory (LRA) at the Aeronautics Technological Institute (ITA).

by enabling the behavior of the real drones to be assessed prior to experimental validation.

The remainder of this paper is organized as follows. Section 2 describes the proposed experimental testbed. Section 3 presents the experimental demonstration of the proposed mixed-reality testbed in three representative scenarios. Finally, Section 4 concludes the paper.

2. SETUP DESCRIPTION

This section describes the proposed mixed-reality indoor testbed for multi-drone experiments involving real and simulated drones. The setup integrates Crazyflie quadcopters, an OptiTrack motion capture system to provide pose measurements for the real drones, and modules developed in MATLAB, Python, and Unity 3D.

The remainder of this section is organized as follows. Subsection 2.1 introduces the Crazyflie platform. Subsection 2.2 briefly describes the OptiTrack motion capture system. Finally, Subsection 2.3 presents the proposed mixed-reality experimental testbed.

2.1 Crazyflie Drone

The Crazyflie, shown in Figure 1, is a compact open-source quadcopter designed for research and educational applications (Bitcraze, 2013). Its lightweight and modular design make it well suited for indoor experiments and for the rapid prototyping of guidance and control algorithms (Giernacki et al., 2017). The platform includes onboard sensors, such as an accelerometer and a gyroscope, and supports integration with external positioning systems, including motion capture systems. In addition, the Crazyflie provides onboard low-level controllers for stabilization and command tracking.

Communication between the Crazyflie and the central computer is established through the Crazyradio dongle (BitCraze, 2013), which enables wireless command transmission and data exchange during the experiments.

2.2 OptiTrack Motion Capture System

The proposed setup employs the OptiTrack motion capture system (OptiTrack, 1996), which consists of a network of infrared cameras capable of detecting reflective markers attached to the tracked drone, as shown in Figure 1. This system provides accurate position and attitude measurements at high sampling rates, which makes it suitable for feedback-based guidance, navigation, and control applications (Soria et al., 2022).

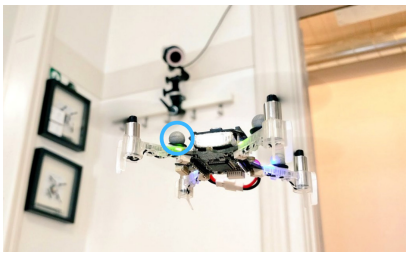


Figure 1. Crazyflie drone equipped with reflective markers.

The data captured by the cameras are processed by the Motive software, which estimates the rigid-body poses and streams them through the local network. These measurements are then received by the central computer running the experimental testbed.

2.3 Mixed-Reality Experimental Testbed

The proposed testbed architecture comprises Crazyflie quadcopters, OptiTrack motion capture cameras, and specific modules developed in MATLAB, Python, and Unity 3D. The pose estimates of the Crazyflies are provided in real time by the motion capture system and streamed to the MATLAB module over the local network.

The MATLAB module runs in soft real time and is the central coordinator of the testbed, enabling configuration of the experimental setup, implementation of guidance and control algorithms, and dynamic simulation of the drones and obstacles. To improve simulation fidelity, the drone dynamics are modeled based on the Crazyflie dynamic model. Moreover, this module enables the coexistence of real and simulated drones within the same experimental environment, which constitutes the main contribution of the setup. This is achieved by acquiring the motion capture pose estimates for the real drones and sending their computed guidance commands, along with their position estimates for indoor localization, to a communication module developed in Python via UDP. This module uses the Crazyswarm library (Preiss et al., 2017) and communicates with the Crazyflie radio dongle to transmit the guidance commands and position feedback to the Crazyflies. To provide the mixed-reality capability, the MATLAB module transmits via UDP the poses of all the drones and obstacles to a photorealistic three-dimensional visualizer developed in Unity 3D. This architecture supports three modes of operation: a pure simulation mode, in which all entities are simulated; an experimental mode, which integrates real drones with virtual obstacles; and a mixed mode, which combines both real and simulated drones alongside virtual obstacles within the same environment.

Figure 2 shows the block diagram of the proposed testbed, while Figure 3 illustrates the single-computer implementation of the mixed-reality setup. On the one hand, it is worth noting that the proposed MATLAB architecture can be readily adapted to incorporate different algorithms, such as those for decision-making and state estimation. On the other hand, the proposed architecture can be distributed across multiple machines on the same local network, as the Motive software, MATLAB module, Python communication script, and Unity 3D visualizer can each run on separate computers. However, the proposed testbed does not support distributing the MATLAB and Python modules across different machines, although this can be achieved with appropriate adaptations to the modules and their communication interfaces.

3. DEMONSTRATION

This section presents three demonstrations illustrating the proposed mixed-reality testbed in differents scenarios involving real and simulated drones. The considered cases are: (i) a fully simulated scenario with three drones, (ii)

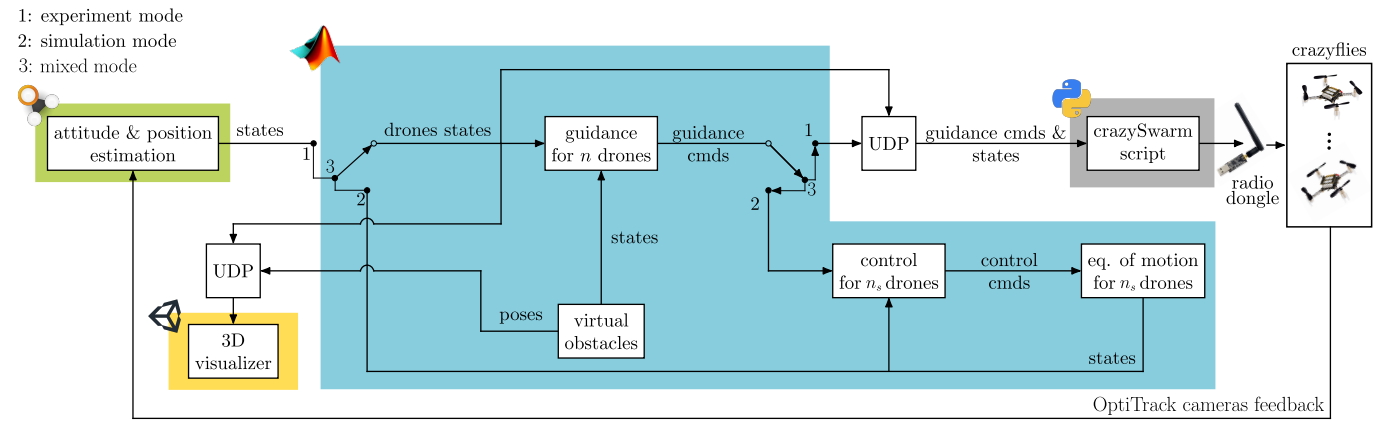


Figure 2. Block diagram of the proposed mixed-reality experimental setup. $n \in \mathbb{Z}_{\geq 0}$ denotes the total number of drones and $n_s \in \mathbb{Z}_{\geq 0}$ denotes the number of simulated drones.

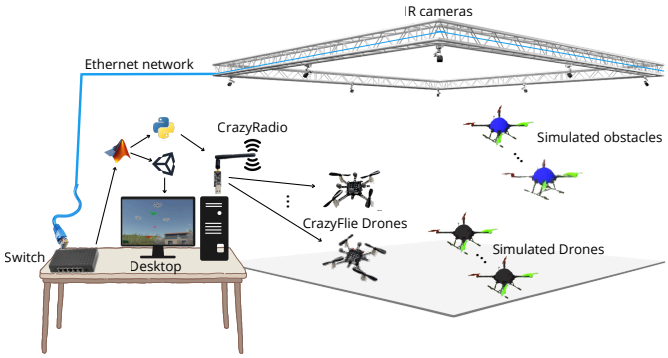


Figure 3. Illustration of the mixed-reality experimental setup.

a hybrid scenario with one real drone and one simulated drone, and (iii) a hybrid scenario with two real drones and one simulated drone. The proposed experimental setup enables the evaluation of different guidance, control, and planning methods, such as the optimal trajectory planning approach in (Cruz et al., 2025) and the collision-free guidance method adopted in the present demonstrations (Ricardo Jr and Santos, 2023a,b).

Although only these configurations are presented, the proposed apparatus is not restricted to them and can support larger numbers of simulated and real agents. In practice, the main limitation for real drones is the available physical space of the indoor arena, which in our setup is $2 \times 2 \times 2$ m.

A video showing the three demonstrations is available at: `https://youtu.be/Pvgecr14T_c`.

3.1 Fully Simulated Scenario

The first demonstration considers a fully simulated scenario with three drones. In this case, all drone states are propagated in MATLAB using the implemented Crazyflie dynamics and equations of motion. The three agents operate within the same simulated environment and execute the trajectories under the proposed testbed.

Figure 4 shows the resulting three-dimensional trajectories. The color variation along each path represents the time evolution. The figure illustrates that the drones complete the maneuver in a coordinated manner, demonstrating the use of the apparatus in a purely simulated scenario.

To provide additional insight into the system behavior, Figure 5 presents the position and velocity components of drone 1 along each axis together with the corresponding command references. These results show the consistency between the generated commands and the simulated closed-loop response.

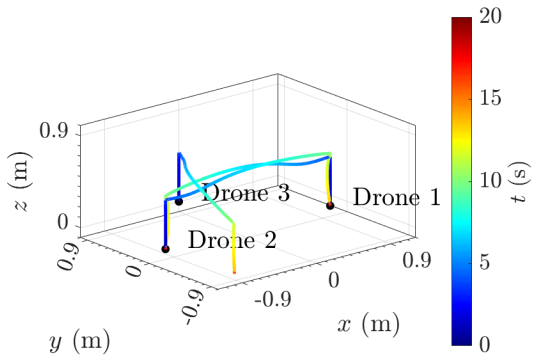
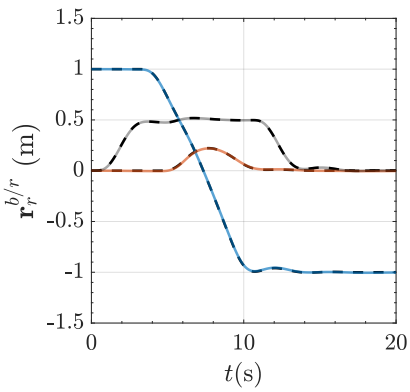


Figure 4. Three-dimensional trajectories of the three simulated drones. The color variation along each trajectory indicates the time evolution. In the figure, • denotes the initial position of each drone, and —— denotes its trajectory.

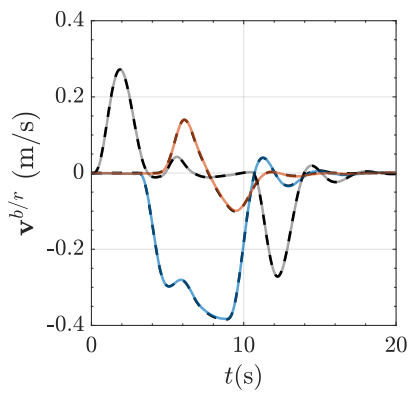
3.2 Hybrid Scenario With One Real and One Simulated Drone

The second demonstration considers a hybrid scenario with one real drone and one simulated drone. The pose of the real drone, including its initial position, is obtained from the OptiTrack motion capture system, while the state of the simulated drone is propagated in MATLAB using the implemented dynamic model. Both agents are handled within the same computational apparatus and share the same mixed-reality environment.

Figure 6 shows the resulting three-dimensional trajectories. The real drone evolves according to the motion capture measurements, while the simulated drone follows



(a) Position



(b) Velocity

Figure 5. Position and velocity components of drone 1. In both plots, —— x axis, —— y axis, —— z axis, and --- command reference.

the numerical model implemented in the central computer. This experiment illustrates the integration of physical and virtual agents operating simultaneously within the same scenario.

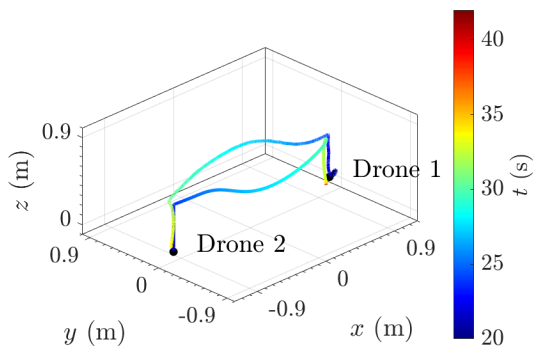


Figure 6. Three-dimensional trajectories of the hybrid scenario with one real drone and one simulated drone. The color variation along each trajectory indicates the time evolution. In the figure, • denotes the initial position of each drone, and —— denotes its trajectory.

3.3 Hybrid Scenario With Two Real and One Simulated Drone

The third demonstration extends the previous hybrid experiment by considering two real drones and one simulated drone. As before, the poses of the real drones are obtained from the OptiTrack system, while the simulated drone state is propagated in MATLAB. All agents are represented within the same workspace and processed by the supervisory mixed-reality testbed.

Figure 7 presents the resulting three-dimensional trajectories. The experiment demonstrates that the proposed testbed can support simultaneous coexistence operation involving multiple physical drones interacting with simulated agents within the same scenario.

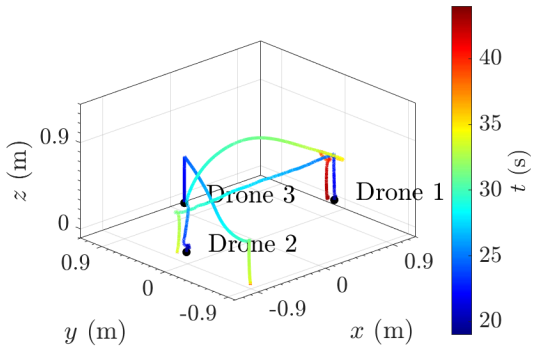


Figure 7. Three-dimensional trajectories of the hybrid scenario with two real drones and one simulated drone. The color variation along each trajectory indicates the time evolution. In the figure, • denotes the initial position of each drone, and —— denotes its trajectory.

4. CONCLUSIONS

This paper has proposed a mixed-reality indoor experimental testbed for multi-drone experiments. Its main contribution lies in extending a previously developed mixed-reality arena to support the coexistence of simulated and real drones within the same experimental environment. The proposed testbed architecture integrates Crazyflie drones, an OptiTrack motion capture system for real-time pose feedback, a Python-based communication interface for the Crazyflie platform, an object-oriented MATLAB module implementing guidance, control, and simulation routines based on the Crazyflie dynamic model, and a photorealistic three-dimensional visualization environment developed in Unity 3D. The coexistence of real and simulated drones is enabled by the MATLAB module, which acts as the central coordinator, properly managing the interfaces among all system components.

The testbed has been validated through three representative scenarios: a fully simulated experiment with three drones, a hybrid setup with one real and one simulated drone, and a hybrid configuration with two real and one simulated drone. These experiments have demonstrated the feasibility and flexibility of the proposed testbed for the development and evaluation of multi-agent guidance and control strategies, while allowing for a reduction in the

number of physical platforms required. This is particularly relevant as it enables experimental validation even when hardware availability is limited.

Future work will focus on distributing the MATLAB and Python modules across multiple machines. This approach is expected to improve scalability and modularity, while reducing dependence on a centralized computational architecture for these modules.

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